Rodrigo Patrício

CYBERSECURITY & DISTRIBUTED SYSTEMS ENGINEER

Portalegre, Portugal • rudrapatricio@gmail.com • +351 964 231 467 • github.com/Nanochad • linkedin.com/in/rodrigo-patrício

PROFILE

MSc Computer Science student at NOVA FCT, specialising in **Cybersecurity** and **Distributed Systems**, after a completed BSc in Informatics Engineering. I like building things at the harder end of computer science: **applied-cryptography** and **distributed-systems** projects ranging from homomorphic-encrypted databases to a Paxos and Raft implementation. I also have C++ development experience and work across AWS, Azure and GCP. EU citizen, fluent in English (C1/C2), open to relocation.

EDUCATION

MSc Computer Science, NOVA School of Science & Technology (FCT NOVA) — Sep 2025 – Jun 2027 (expected)
Specialisation in Distributed Systems & Cybersecurity · Lisbon, Portugal

BSc Informatics Engineering, NOVA School of Science & Technology (FCT NOVA) — Sep 2021 – Jun 2025
Computer Science & Engineering · Lisbon, Portugal

EXPERIENCE

Software Developer, STAB VIDA — Oct 2024 – Feb 2025
Lisbon, Portugal

- Built new features in **C++** for *DoctorVida Education*, a microcontroller-based spectrophotometer device used in educational settings.
- Worked close to the hardware, handling sensor data within tight embedded constraints, as part of a small product team that shipped and tested features.

PROJECTS

Homomorphic-Encrypted Employee Database — APPLIED CRYPTOGRAPHY
Built a database that computes directly over encrypted records using **Paillier** additively-homomorphic encryption and **Boldyreva order-preserving encryption (OPE)**, so range queries and aggregation run without ever decrypting the underlying data.

State Machine Replication with Paxos & Raft — DISTRIBUTED SYSTEMS
Implemented the **Paxos** and **Raft** consensus algorithms to provide fault-tolerant, strongly consistent state-machine replication across distributed nodes.

Encrypted Streaming Server — SECURITY / NETWORKING
Wrote a media streaming server that protects content in transit, keeping delivery between server and clients secure.

Lisbon Public-Transport Graph — NEO4J / GRAPH DB
Modelled Lisbon's public-transport network as a **Neo4j** graph database, answering route and connectivity questions with Cypher.

TECHNICAL SKILLS

Languages	Python · Java · C++ · SQL · Cypher
Security & Crypto	Homomorphic (Paillier) and order-preserving (Boldyreva) encryption · Cloud security · Linux · Risk management
Distributed Systems	Paxos · Raft · State machine replication · Fault tolerance
Cloud & Data	AWS · Microsoft Azure · GCP · Kubernetes · Azure Cosmos DB · Neo4j graph databases

CERTIFICATIONS

- Foundations of Cybersecurity (Google)
- Play It Safe: Manage Security Risks (Google)
- Tools of the Trade: Linux and SQL (Google)
- Cybersecurity Foundation (Palo Alto Networks)
- Cloud Technical Essentials (AWS)

LANGUAGES

Portuguese: Native | **English**: Proficient (C1 / C2)